

Case Study

Business Context :

A large-scale performance marketing platform runs millions of creatives across multiple geographies, formats, and platforms (Meta, Google, TikTok, OTT). Creative fatigue, delayed feedback loops, and attribution noise make creative performance highly volatile and hard to predict. The company wants to move from reactive creative testing to a proactive, self-improving creative intelligence engine.

Core Problem

Creative performance is non-stationary, context-dependent, and platform-specific. Historical CTR alone is insufficient due to:

- Targeting shifts
- Creative fatigue and novelty decay
- Platform-level bias in delivery

Deliverables (3 to 5 Pager PDF Document) :

- 1) High-level system architecture
- 2) Data schema and feature strategy
- 3) Modelling approach (with trade-offs)

Things to be covered under the deliverables :

Design an end-to-end AI system that can understand, predict, generate, and continuously optimise ad creatives before and after launch, while operating under real-world data and infra constraints.

System Requirements

1. Multimodal Creative Understanding

Build a vision + language system that:

- Parses existing static images, short-form videos, and long-form creatives
- Extracts semantic, visual, and temporal features,

2. Pre-Launch Performance Prediction

Design a model that:

- Predicts expected CTR distribution (not point estimate) before launch
- Decomposes predicted CTR into interpretable creative drivers

3. Causal Creative Attribution

Move beyond correlation:

- Identify which creative elements causally impact CTR

Separate creative impact from: Targeting, Bid , Seasonality

4.LLM-Guided Creative Optimization

Design an LLM-driven system that recommends on

- Copy rewrite
- Color palette changes
- CTA restructuring
- Frame sequencing changes
- Generates structured prompts for creative regeneration

5.Human-in-the-Loop Feedback & Autonomous Creative Generation

Incorporate:

- Marketer feedback (accept/reject/edit suggestions)
- Performance outcomes post-deployment
- Creative fatigue signals over time

Generate a new static or video creation from Steps 3, 4, and Human feedback.